

MY PROJECTS - INTERVIEW PREPARATION GUIDE

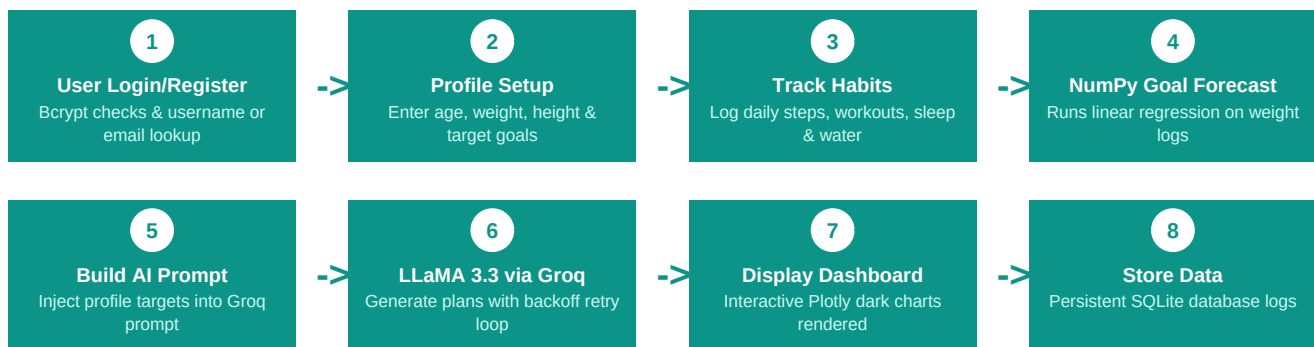
K. GNANESWAR | Project 1: AI Gap-Chatbot - Project 2: AI Fitness App - Project 3: AirCanvas

PROJECT 2 - AI-Powered Fitness Application (FitAI)

Abstract - What to Say in Interview

FitAI is a premium, full-stack health tracking and AI-driven coaching web application. Users register and log in securely with passwords hashed using bcrypt. They can log workouts, water, weight, steps, sleep, and calorie balances. The system uses NumPy to perform linear regression calculations on historical weight logs to predict the target date a user will achieve their fitness goal. Plotly generates 10 dark-themed interactive charts to visualize progress, and the application constructs rich data prompts to call LLaMA 3.3 70B via the Groq API to serve custom training programs and meal plans. Data is stored in a local, persistent SQLite database, and the application is deployed on PythonAnywhere.

System Architecture



Tech Stack & Library Roles

Library / Tool	Role in Project
Flask (Python)	Backend framework managing routing blueprints and DB context
SQLite	Zero-configuration database storing profiles and daily tracking metrics
bcrypt	Blowfish cryptographic library hashing user passwords securely
Groq API (Llama 3.3)	AI completion engine generating meal plans, workouts, and chats
NumPy	Performs polyfit linear regression calculations to forecast weight dates
Pandas	Aggregates history databases and exports records cleanly into CSV files
Plotly.js	Generates client-side hardware-accelerated dark dashboard charts
python-dotenv	Secures API keys and session secrets loaded on startup

Key Interview Q&A

Q: Did you train the AI model?

-> No. I used a pre-trained LLaMA 3.3 70B model via Groq API. My focus was on building the tracking pipeline, prompt context injection, and user interfaces around it.

Q: What is your contribution?

-> I built the full-stack system: setup SQLite schemas, implemented bcrypt password security and username/email authentication, developed the NumPy regression forecaster, compiled Plotly charts, and designed the glassmorphism CSS

layout.

Q: Why SQLite instead of SQL Server?

-> SQLite is local and file-based. Since users query only their own logs, concurrency is low, making SQLite's locking model highly efficient, zero-config, and ideal for persistent storage on PythonAnywhere.

Q: How does the NumPy linear regression weight predictor work?

-> It extracts log coordinates (date, weight) and uses `np.polyfit` to calculate the trendline slope and intercept ($y = mx + c$). Solving for the user's goal weight calculates the estimated target date.

Q: How does the AI chatbot maintain conversation history context?

-> It fetches the user's last 20 messages from the `chat_history` database, arranges them chronologically as a history log, and appends it to the API message array sent to Groq.